

Software Requirements Specification for Fraud Detection Management Dashboard

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Table of Contents

Table of Contents

Revision History

1. Introduction

- 1.1 Purpose
- 1.2 Document Conventions
- 1.3 Intended Audience and Reading Suggestions
- 1.4 Product Scope
- 1.5 References

2. Overall Description

- 2.1 Product Perspective
- 2.2 Product Functions
- 2.3 User Classes and Characteristics
- 2.4 Operating Environment
- 2.5 Design and Implementation Constraints
- 2.6 User Documentation
- 2.7 Assumptions and Dependencies

3. External Interface Requirements

- 3.1 User Interfaces
- 3.2 Hardware Interfaces
- 3.3 Software Interfaces
- 3.4 Communications Interfaces

4. System Features

- 4.1 User Authentication (FR-01)
- 4.2 Transaction Monitoring (FR-02)
- 4.3 Fraud Detection (FR-03)
- 4.4 Risk Score Generation (FR-04)
- 4.5 Transaction Classification (FR-05)
- 4.6 Fraud Alerts (FR-06)
- 4.7 Transaction History (FR-07)
- 4.8 Fraud Investigation (FR-08)
- 4.9 Fraud Case Management (FR-09)
- 4.10 Reports and Analytics (FR-10)

4.11 User Management (FR-11)

4.12 Detection Rule Management (FR-12)

4.13 Role-Based Access Control (FR-13)

5. Other Nonfunctional Requirements

5.1 Performance Requirements

5.2 Safety Requirements

5.3 Security Requirements

5.4 Software Quality Attributes

5.5 Business Rules

6. Other Requirements

Appendix A: Glossary

Appendix B: Analysis Models

Revision History

Name	Date	Reason For Changes	Version
Akshay Pratap Singh et al.	10/09/26	Initial Release of Software Requirements Specification	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to detail the requirements for the "Fraud Detection Management Dashboard" (Version 1.0). This document serves as a guideline for developers, stakeholders, and testers to ensure the system is designed, developed, and implemented according to the specified requirements.

1.2 Document Conventions

This document follows standard IEEE typographical conventions. Functional requirements are tagged as FR-01 through FR-13, and non-functional requirements are tagged under NFR classifications. Priority levels (High, Medium, Low) are explicitly stated for each functional feature.

1.3 Intended Audience and Reading Suggestions

This document is intended for software developers, project managers, fraud analysts, system administrators, and academic evaluators involved in the design and assessment of the project. Developers and system architects should focus on Sections 2, 3, and 4, while testers and quality managers should review Section 4 and Appendix B.

1.4 Product Scope

The Fraud Detection Management Dashboard is designed to monitor financial transactions, detect potentially fraudulent activities, calculate risk scores, classify transactions, generate alerts, and assist fraud analysts in managing cases. It aims to enhance transaction security and streamline fraud investigation processes.

1.5 References

- [Reference 1: Academic Textbook on Software Engineering]
- [Reference 2: Research Paper on Fraud Detection Techniques]
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

2. Overall Description

2.1 Product Perspective

The Fraud Detection Management Dashboard is a standalone system that interfaces with existing banking and payment systems. It leverages both rule-based and machine learning techniques to detect and manage fraudulent activities.

2.2 Product Functions

- Monitor transactions in real-time
- Detect and classify potential fraud
- Generate risk scores
- Issue alerts and notifications
- Manage fraud investigations and cases
- Generate reports and analytics

2.3 User Classes and Characteristics

- **Customer/User:** Individuals performing transactions.
- **Fraud Analyst:** Specialists investigating suspicious activities.
- **Manager:** Oversees fraud management operations.
- **System Administrator:** Maintains system functionality and security.
- **Bank/Payment System:** Provides transaction data for analysis.

2.4 Operating Environment

The system will operate on secure servers with internet access, compatible with modern web browsers for user access.

2.5 Design and Implementation Constraints

- Must comply with legal and regulatory standards for data privacy and security.
- Must integrate with existing banking systems via standard APIs.

2.6 User Documentation

User manuals, administrator operational guides, and inline dashboard help documentation will be delivered alongside the software.

2.7 Assumptions and Dependencies

- Reliable access to transaction data from banking systems.
- Use of up-to-date fraud detection algorithms and models.

3. External Interface Requirements

3.1 User Interfaces

Web-based interface compatible with modern browsers providing real-time data visualizers, case review screens, and administrative control panels.

3.2 Hardware Interfaces

Operates on standard server hardware and cloud infrastructure supporting TCP/IP network interfaces.

3.3 Software Interfaces

Integrates with existing banking APIs and databases for seamless real-time data exchange.

3.4 Communications Interfaces

Secure internet connection using encrypted protocols (HTTPS/TLS) for all data transmissions.

4. System Features

4.1 User Authentication (FR-01)

4.1.1 Description and Priority

Allows users to securely register and log in to the system. Priority: High.

4.1.2 Stimulus/Response Sequences

User enters credentials → System authenticates against stored data → Access granted or denied.

4.1.3 Functional Requirements

REQ-1 (FR-01): The system must authenticate users before granting access. Input: User credentials. Output: Access granted or denied.

4.2 Transaction Monitoring (FR-02)

4.2.1 Description and Priority

Continuously monitors incoming transactions in real-time. Priority: High.

4.2.2 Stimulus/Response Sequences

Transaction data received from banking system → System processes continuously → Transaction logged.

4.2.3 Functional Requirements

REQ-2 (FR-02): Monitor all incoming transactions in real-time. Input: Transaction data. Output: Logged transactions.

4.3 Fraud Detection (FR-03)

4.3.1 Description and Priority

Identifies potentially fraudulent transactions using ML models and rules. Priority: High.

4.3.2 Stimulus/Response Sequences

Transaction data available → Apply detection rules/ML model → Suspicious activity flagged.

4.3.3 Functional Requirements

REQ-3 (FR-03): Detect potentially fraudulent transactions. Input: Transaction data. Output: transactions.

Flagged

4.4 Risk Score Generation (FR-04)

4.4.1 Description and Priority

Calculates a risk score for each transaction. Priority: Medium.

4.4.2 Stimulus/Response Sequences

Transaction details available → Analyze transaction parameters → Assign risk score.

4.4.3 Functional Requirements

REQ-4 (FR-04): Generate a risk score for each transaction. Input: Transaction parameters. Output: Risk score.

4.5 Transaction Classification (FR-05)

4.5.1 Description and Priority

Classifies transactions based on risk assessment. Priority: Medium.

4.5.2 Stimulus/Response Sequences

Risk score available → Execute classification logic (Low risk → Approve, Medium risk → Verify, High risk → Block) → Status updated.

4.5.3 Functional Requirements

REQ-5 (FR-05): Classify transactions based on risk score. Input: Risk score. Output: Transaction status.

4.6 Fraud Alerts (FR-06)

4.6.1 Description and Priority

Generates alerts for suspicious transactions. Priority: High.

4.6.2 Stimulus/Response Sequences

Suspicious transaction detected → Alert generated → Sent to fraud analysts and managers.

4.6.3 Functional Requirements

REQ-6 (FR-06): Generate alerts for suspicious transactions. Input: Suspicious transaction data. Output: Alerts.

4.7 Transaction History (FR-07)

4.7.1 Description and Priority

Maintains a history of all processed transactions. Priority: Medium.

4.7.2 Stimulus/Response Sequences

Transactions processed → Store transaction data → Accessible for historical review.

4.7.3 Functional Requirements

REQ-7 (FR-07): Store transaction history for analysis and review. Input: Transaction data. Output: Historical records.

4.8 Fraud Investigation (FR-08)

4.8.1 Description and Priority

Provides tools for in-depth investigation of flagged transactions. Priority: Medium.

4.8.2 Stimulus/Response Sequences

Transaction flagged → Analyst accesses details → Investigation report generated.

4.8.3 Functional Requirements

REQ-8 (FR-08): Provide tools for fraud investigation. Input: Flagged transaction data. Output: Investigation report.

4.9 Fraud Case Management (FR-09)

4.9.1 Description and Priority

Manages cases of confirmed fraud. Priority: High.

4.9.2 Stimulus/Response Sequences

Fraud confirmed → Create new case → Assign to analyst → Track progress to resolution.

4.9.3 Functional Requirements

REQ-9 (FR-09): Manage fraud cases effectively. Input: Confirmed fraud data. Output: Case records.

4.10 Reports and Analytics (FR-10)

4.10.1 Description and Priority

Generates reports and analytics on fraud trends. Priority: Medium.

4.10.2 Stimulus/Response Sequences

Select report parameters → Analyze transaction & fraud data → Generate report.

4.10.3 Functional Requirements

REQ-10 (FR-10): Generate fraud trend reports. Input: Transaction and fraud data. Output: Reports.

4.11 User Management (FR-11)

4.11.1 Description and Priority

Manages user accounts and access levels. Priority: Medium.

4.11.2 Stimulus/Response Sequences

Admin opens user management → Create/update/delete accounts → Assign roles.

4.11.3 Functional Requirements

REQ-11 (FR-11): Manage user roles and permissions. Input: User data. Output: User records.

4.12 Detection Rule Management (FR-12)

4.12.1 Description and Priority

Manages predefined fraud detection rules. Priority: Medium.

4.12.2 Stimulus/Response Sequences

Analyst/Admin selects rule configuration → Add/modify/delete rules → Save updated logic.

4.12.3 Functional Requirements

REQ-12 (FR-12): Manage fraud detection rules. Input: Rule data. Output: Updated rules.

4.13 Role-Based Access Control (FR-13)

4.13.1 Description and Priority

Ensures access control based on user roles. Priority: High.

4.13.2 Stimulus/Response Sequences

User requests resource → System verifies role permissions → Enforce access/deny.

4.13.3 Functional Requirements

REQ-13 (FR-13): Enforce role-based access control. Input: User role data. Output: Secured access.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system must process transactions in real-time with minimal latency (<200ms processing window).

5.2 Safety Requirements

The system must prevent data loss during power/network failures and ensure continuous system transaction state integrity.

5.3 Security Requirements

Ensured data and system security through encryption (in transit and at rest), multi-factor authentication, audit logs, protection of sensitive user info, and unauthorized access prevention.

5.4 Software Quality Attributes

- **Reliability:** Minimal downtime, fault tolerance.
- **Scalability:** Support increasing transaction volumes without performance degradation.
- **Availability:** Ensure 99.9% uptime.
- **Usability:** Intuitive user interface for all user roles.
- **Maintainability:** Easy system updates and modular maintenance.
- **Accuracy:** High accuracy in fraud detection and risk scoring.
- **Privacy:** Protect user data privacy in compliance with regulations.

5.5 Business Rules

Only certified Fraud Analysts and Managers can override automated risk classification blocks or close confirmed fraud cases.

6. Other Requirements

Error Handling: Validate invalid transaction data and notify users. Implement default safety fallbacks for missing transaction details.

Appendix A: Glossary

- **SRS:** Software Requirements Specification
- **ML:** Machine Learning
- **UI:** User Interface
- **API:** Application Programming Interface

Appendix B: Analysis Models

B.1 System Workflow

1. Transaction Received
2. Transaction Validation
3. Fraud Detection
4. Risk Analysis
5. Risk Score Generation
6. Transaction Classification
7. Approve / Verify / Flag / Block Action

- 8. Alert Generation
- 9. Investigation
- 10. Case Management
- 11. Resolution
- 12. Reporting

B.2 Decision Table

Conditions / Actions	R1	R2	R3	R4	R5
High Amount	T	F T	F F	T	F F
Unusual Location	F	F F	T F	T	F T
New Device	F	F	T	T	T
High Frequency	F			T F	
Unusual Behavior	F				
Approve	T				
Request Verification		T			
Flag Suspicious			T		
Block				T	
Generate Alert				T	T
Create Case					T

B.3 Data Dictionary

Entity	Attribute	Data Type	Description
User	UserID	String	Unique identifier for the user
	UserName	String	Name of the user
Transaction	TransactionID	String	Unique identifier for a transaction
	Amount	Decimal	Amount of the transaction
Fraud Alert	AlertID	Integer	Unique identifier for a fraud alert
Fraud Case	CaseID	String	Unique identifier for a fraud case
Risk Score	Score	String	Calculated risk score for a transaction
Detection Rule	RuleID	String	Identifier for a fraud detection rule
Report	ReportID	String	Unique identifier for a generated report

B.4 Requirement Traceability Matrix

Requirement ID	Requirement	Actor/Stakeholder	Related Module	Priority
FR-01	User Authentication	Customer, Administrator	User Authentication	High
FR-02	Transaction Monitoring	Bank/Payment System	Transaction Monitoring	High
FR-03	Fraud Detection	Fraud Analyst	Fraud Detection	High
FR-04	Risk Score Generation	Fraud Analyst	Risk Score Generation	Medium
FR-05	Transaction Classification	Fraud Analyst	Transaction Classification	Medium
NFR-01	Security Features	All Users	Security	High